OPERATING SYSTEMS [2 1 0 3]

[Revised Credit System]

(Effective from the academic year 2023-2024)

SEMESTER – V

Subject Code	**CSE 3153**	**IA Marks**	**50**
Number of Lecture Hours/Week	**03**	**Exam Marks**	**50**
Total Number of Lecture Hours	**36**	**Exam Hours**	**03**
CREDITS – 03			

Course objectives: This course will enable students to

- To understand the fundamental concepts with the basic privileges, structure, protection and security of the operating system.
- Understand the fundamental concepts of the basic privileges, structure and process scheduling of the operating system.
- Comprehend the concepts of deadlock, memory management, virtual memory and disk structure concepts.
- Know the concepts of the file system and protection of operating system.

Module -1	**Teaching Hours**
INTRODUCTION: What Operating Systems Do, Operating System Structure, Operating System Operations, Process Management, Memory Management, Storage Management. **OPERATING SYSTEM STRUCTURE:** Operating System Services, User and Operating System Interface, System Calls, Types of System Calls, System Programs, Operating System Structure, System Boot. **PROCESS CONCEPT:** Overview, Process Scheduling, Operations on Processes, Interprocess Communication. **THREADS:**	**10 Hours**

Overview, Multithreaded Models, Thread Libraries. **Text Book 1:** Chapter: 1: 1.1,1.4-1.8, Chapter: 2: 2.1– 2.5, 2.6,2.7.1-2.7.3, 2.10, Chapter: 3: 3.1-3.4, Chapter: 4: 4.1, 4.3-4.4	
Module -2	
PROCESS SCHEDULING: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Thread Scheduling **SYNCHRONIZATION:** Background, The Critical Section Problem, Peterson's Solution, Synchronization Hardware, Mutex Locks, Semaphores. **DEADLOCKS:** System Model, Deadlock, Characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock Avoidance. **Text Book 1:** Chapter 5: 5.1-5.4, Chapter: 6: 6.1-6.6, Chapter: 7: 7.1 -7.5	**10 Hours**
Module – 3	
MEMORY MANAGEMENT STRATEGIES: Logical Versus Physical Address Space, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table. **VIRTUAL-MEMORY MANAGEMENT:** Background, Demand Paging, Copy-On-Write, Page Replacement, Allocation of Frames, Thrashing. **MASS-STORAGE STRUCTURE:** Disk Structure, Disk Scheduling, Swap-Space Management	**10 Hours**

Text Book 1: Chapter 8: 8.1.3,8.2-8.6, Chapter 9: 9.1-9.6, Chapter 12: 12.2 12.4,12.6	
Module - 4	
FILE-SYSTEM: File Concept, Access Methods, Directory and Disk Structure, File System Mounting **SYSTEM PROTECTION:** Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix Implementation of Access Matrix. **Text Book 1:** Chapter 10: 10.1- 10.4, Chapter 14 :14.1- 14.5	**6 Hours**

Course outcomes:

After studying this course, students will be able to:

1. Outline the concepts of threads and process.
2. Interpret the different types of process synchronization techniques
3. Analyze different CPU scheduling and deadlock algorithms
4. Examine the main memory and the virtual memory concepts
5. Summarize the file systems and protection related to a general operating system

Text Books:

1. A. Silberschatz, P. B. Galvin and G. Gagne, *Operating System Concepts*, (9e), Wiley and Sons (Asia) Pte Ltd, 2016.

Reference Books:

1. D.M. Dhamdhere, *Operating Systems A Concept based Approach,(2e),* Tata McGraw-Hill 2006

2. Deitel & Deitel , *Operating systems*, *(3e),* Pearson Education, India. 2018.

3. Neil Matthew, Richard Stones, Beginning Linux Programing(4e), Wiley, 2007

4. Andrew S Tanenbaum, Herbert Bos, *Modern Operating systems*. Pearson Education India, 2018